Federated Full-Envelope LPV Identification from Incomplete Local Coverage

***Abstract*—Vehicles may supply calibration data from narrow speed regions while the fleet collectively covers a wider interval. Pooling can recover missing scheduling information but mix incompatible dynamics. We study federated linear-parameter-varying (LPV) identification under these joint limitations. Clients exchange additive regression statistics and assignment scores to learn shared group maps. Across ten nonlinear vehicle fleets with family labels withheld, grouped fitting reduces mean paired full-envelope matrix error by 56.20% relative to development-tuned Smooth Local and 23.74% relative to Global. Mean error at locally unobserved speeds is 3.527%, versus 9.458% and 4.388%, respectively. The distributed fit reproduces centralized grouping. Benefits persist under alternative entry scaling but are not uniform across entries or clients. The benchmark uses controlled equilibrium-centered calibration with available states, exact current regressors, and noisy next-state targets. Fresh stress tests expose fleet-wide coverage failure and loss of grouping gains at lower calibration signal-to-noise. Speed-scheduled linear-quadratic-integral (LQI) tests show modest gains over Global; the tested Smooth Local prior yields infeasible responses for some clients under the frozen controller protocol.**

***Index Terms*—federated identification, LPV systems, incomplete operating coverage, heterogeneous vehicles**

I. Introduction

Model-based vehicle control requires dynamics models that remain useful as operating conditions change. For lateral motion, longitudinal speed affects the relationship between steering, sideslip, and yaw rate. A model identified near one operating point may therefore be inadequate elsewhere. LPV models represent this dependence explicitly, making identification across the intended speed range a central requirement [1], [2].

Obtaining that coverage separately for every vehicle is a demanding calibration task. A vehicle may provide measurements over only a narrow part of its intended operating range. Those measurements constrain the observed response, but a flexible speed-dependent model can remain undetermined outside that region. Repeating measurements at the same speeds reduces noise without resolving missing scheduling directions. The resulting problem is therefore incomplete operating information as well as limited sample count.

Fleet collaboration offers a way to fill these gaps: one vehicle's unobserved speeds may be covered by others. Yet differences in mass, inertia, and tire properties make their measurements only approximately compatible. Pooling all vehicles can introduce bias, while separating them too aggressively can leave each group without sufficient speed coverage. What is needed is selective sharing that supplies missing operating information without overwhelming a recipient's dynamics with incompatible data. Crucially, the partners must be chosen from partial observations, before complete local models are available.

We address this need through federated LPV identification. Vehicles retain calibration records locally, exchange additive statistics, and evaluate candidate group maps at their observed speeds. Alternating assignment and regression learns shared maps over the wider interval. The central question is whether groups inferred from partial data can provide both complementary coverage and a useful approximation of their recipients' dynamics. Figure 1 summarizes the procedure.

Existing work supplies important components of this approach. Local LPV identification constructs scheduling-dependent maps from operating-point models [3]. Federated learning provides a framework for collaboration without centralizing records [4], and FedSysID characterizes the effect of dynamical heterogeneity on collaborative identification [5]. IFCA alternates cluster assignment and model optimization [6]. Clustered SysID applies this structure to LTI systems with identical dynamics within each true cluster [7]. IC-SYSID adds incremental clustering and stability promotion [8], while [9] uses uncertainty-aware grouping for vehicle controller design.

Our distinction is the joint requirement of compatibility and complementary scheduling information. Each client's fixed speed support leaves its LPV regression deficient, and shared maps remain approximate within groups. The cited recovery guarantees do not establish recovery in this setting. We specialize the established alternating clustering structure to this problem and make the following contributions:

1) We formulate LPV identification from incomplete local speed coverage in common physical-state coordinates. A conditional rank statement distinguishes information supplied by pooling from extrapolation imposed by a local prior.
2) We implement label-free group regression using additive statistics and local assignment scores, without requiring complete local LPV maps. Agreement with centralized grouping verifies the distributed implementation.
3) We establish controlled vehicle-fleet evidence for the coverage–bias tradeoff through comparisons with local priors and global pooling, learned-group and recipient audits, and coverage and calibration stress tests. Supporting LQI tests assess whether identification gains translate into tracking improvements.

Sections II and III present the identification problem and federated regression. Section IV describes the benchmark. Sections V and VI report identification and LQI tracking results. Section VII discusses the findings and concludes.

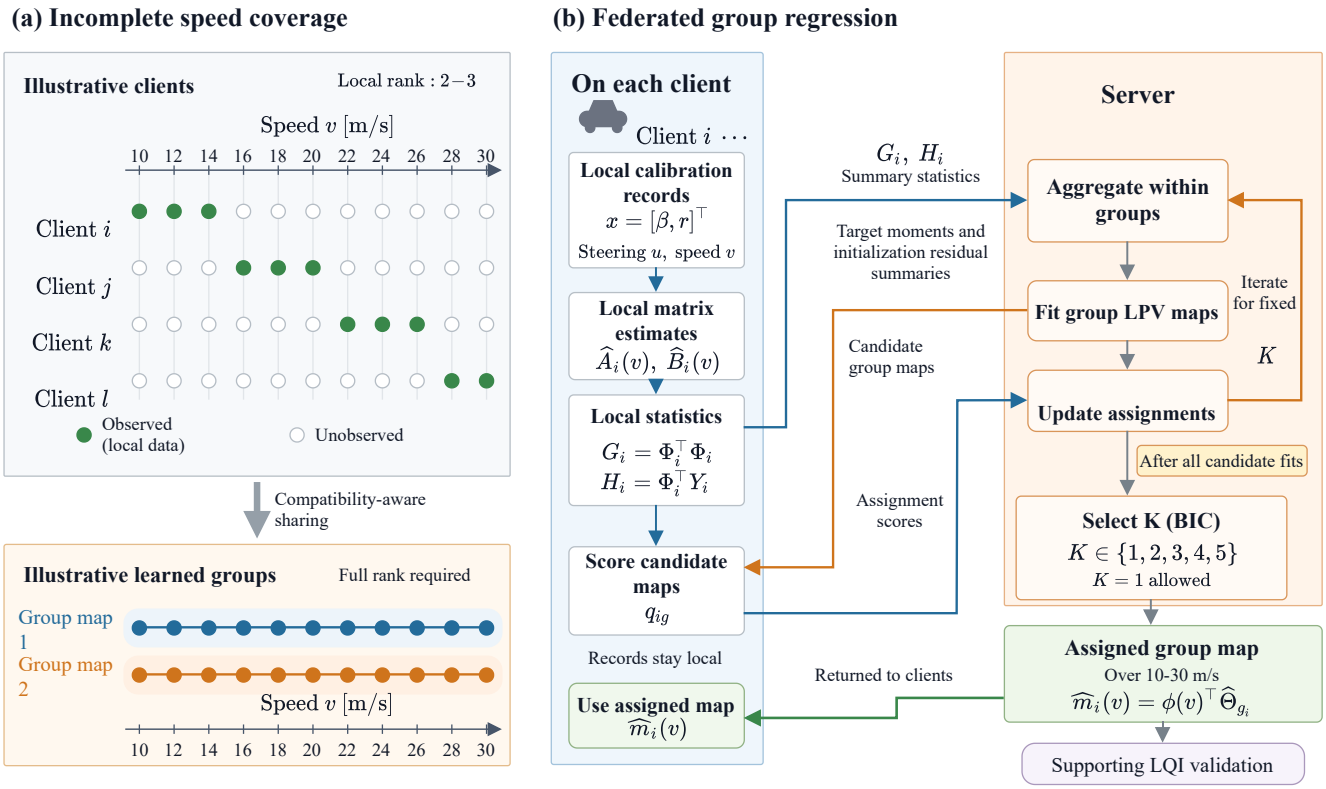


Fig. 1. Compatible federated LPV identification. (a) Illustrative clients and groups. Local speed blocks give scheduling rank two or three for a seven-function basis; sharing can supply missing directions when the pooled group design has rank seven. (b) Clients retain records and matrix targets, exchange additive statistics, residual summaries, and assignment scores, and receive group maps. Fitting and assignment alternate for each candidate K, followed by selection using the Bayesian information criterion (BIC). Available states and equilibrium-centered calibration are assumed. The envelope is 10–30 m/s along the prescribed cornering curve; LQI checks downstream usability.

II. Identification Problem

A. Local calibration information

Consider a fleet of M vehicles indexed by $i \in \{1, \dots, M\}$. The target speed interval is $\mathcal{V} = [v_{\min}, v_{\max}]$, where $0 < v_{\min} < v_{\max}$. Each vehicle has state $x_i = [\beta_i, r_i]^\top \in \mathbb{R}^2$, containing sideslip angle β_i and yaw rate r_i, and steering input $u_i \in \mathbb{R}$. States, steering, and speed are available during calibration. At a fixed speed v, let $x_i^\star(v)$ and $u_i^\star(v)$ denote a prescribed cornering equilibrium. Define deviations $\Delta x_{i,k} = x_{i,k} - x_i^\star(v)$ and $\Delta u_{i,k} = u_{i,k} - u_i^\star(v)$, where k indexes sampled transitions. The local discrete-time approximation is

$$\Delta x_{i,k+1} = A_i(v)\Delta x_{i,k} + B_i(v)\Delta u_{i,k} + e_{i,k}, \tag{1}$$

where $A_i(v) \in \mathbb{R}^{2\times 2}$ and $B_i(v) \in \mathbb{R}^{2\times 1}$ describe perturbation dynamics. The residual $e_{i,k} \in \mathbb{R}^2$ includes nonlinear approximation error and next-state observation noise. Let $m_i(v) \in \mathbb{R}^{1\times 6}$ collect these matrix entries at speed v, in the order $[A_{i,11}, A_{i,12}, A_{i,21}, A_{i,22}, B_{i,1}, B_{i,2}]$.

Vehicle i supplies calibration records at the finite speed set $\mathcal{V}_i = \{v_{i,1}, \dots, v_{i,n_i}\} \subset \mathcal{V}$, where n_i is the number of observed speeds. At each such speed, stack the rows $[\Delta x_{i,k}^\top, \Delta u_{i,k}]$ into $Z_i(v)$ and the corresponding observed next-state deviations into $X_{i,+}(v)$. For $N_i(v)$ records, these matrices have dimensions $N_i(v) \times 3$ and $N_i(v) \times 2$. Ordinary least squares (OLS) estimates the local matrices:

$$\begin{aligned} \widehat{C}_i(v) &= \arg\min_{C \in \mathbb{R}^{3\times 2}} \|X_{i,+}(v) - Z_i(v)C\|_F^2, \\ [\widehat{A}_i(v), \widehat{B}_i(v)] &= \widehat{C}_i(v)^\top. \end{aligned} \tag{2}$$

Here $\|\cdot\|_F$ is the Frobenius norm. We require $\operatorname{rank} Z_i(v) = 3$ for a unique local OLS solution. Vectorizing the estimates in the order of $m_i(v)$ gives $\widetilde{m}_i(v)$. The local identification data are $\mathcal{D}_i = \{(v, \widetilde{m}_i(v)) : v \in \mathcal{V}_i\}$.

B. Incomplete scheduling information and shared maps

Let $\phi(v) \in \mathbb{R}^L$ be a fixed vector of L linearly independent scheduling functions, common to all vehicles. Define $\Phi_i \in \mathbb{R}^{n_i \times L}$ by row ℓ equal to $\phi(v_{i,\ell})^\top$, and $Y_i \in \mathbb{R}^{n_i \times 6}$ by row ℓ equal to $\widetilde{m}_i(v_{i,\ell})$, for $\ell = 1, \dots, n_i$. The setting of interest has

$$\operatorname{rank} \Phi_i < L. \tag{3}$$

Consequently, a nonzero coefficient perturbation $U_i \in \mathbb{R}^{L\times 6}$ can satisfy $\Phi_i U_i = 0$: coefficient maps that differ by U_i agree at every locally observed speed but can disagree elsewhere. More records at those speeds improve the matrix estimates without removing this scheduling ambiguity. Full rank of $Z_i(v)$

and full rank of Φ_i therefore address different requirements: local state/input excitation and speed coverage.

Sharing introduces K group maps, where $g_i \in \{1, \dots, K\}$ is vehicle i's assignment and $\Theta_g \in \mathbb{R}^{L\times 6}$ contains the coefficients of group g. Write

$$m_i(v) = \phi(v)^\top \Theta_{g_i} + d_i(v), \tag{4}$$

where $d_i(v) \in \mathbb{R}^{1\times 6}$ represents basis approximation error and the difference between individual and shared dynamics. Complementary supports can remove missing scheduling directions in a pooled design. However, sharing also couples vehicles through a common approximation. The residual $d_i(v)$ expresses the resulting recipient-dependent mismatch, distinct from noise in the estimated targets Y_i.

C. Identification objective and scope

Given the distributed datasets $\mathcal{D}_i$, we seek a group count K, assignments g_i, and coefficient matrices Θ_g that provide each recipient with a map $\widehat{m}_i(v) = \phi(v)^\top \widehat{\Theta}_{g_i}$ throughout $\mathcal{V}$. The scientific objective is to improve identification at locally unobserved speeds $\mathcal{V} \setminus \mathcal{V}_i$, while limiting the mismatch introduced by pooling heterogeneous vehicles. Those speeds have no local fitting targets. Section III therefore fits observed data and selects groups from their residuals; the evaluation tests whether this procedure improves predictions outside each local support.

Vehicles retain calibration records locally and exchange the statistics and scores defined in Section III. Vehicle parameters, exact evaluation matrices, and family labels do not enter that fitting or assignment interface. The contributing vehicles also receive the maps. We evaluate generalization across speeds within each fleet, with identification repeated for each fresh fleet. Transfer of a fixed model library to new recipients is outside this study. "Full envelope" denotes the target speed interval along the prescribed cornering curve. It does not denote an arbitrary nonlinear state/input domain.

III. Federated Compatible Regression

Using Φ_i and Y_i from Section II, we alternate group-map fitting and vehicle assignment, then select the group count by penalized residual error. These steps use only the locally observed speeds.

A. Additive fitting and conditional identifiability

For a fixed partition, let $\mathcal{I}_g = \{i : g_i = g\}$ be the set of vehicles in group g. Its map minimizes the regularized least-squares objective

$$\widehat{\Theta}_g = \arg\min_{\Theta\in\mathbb{R}^{L\times 6}} \left\{ \sum_{i\in\mathcal{I}_g} \|Y_i - \Phi_i\Theta\|_F^2 + \rho\|\Theta\|_F^2 \right\}, \tag{5}$$

where $\rho = \lambda + \epsilon$, with ridge strength $\lambda = 10^{-8}$ and numerical diagonal adjustment $\epsilon = 10^{-12}$. Each matrix target has equal weight; local estimation covariances are not used, including in the sensitivity study. Client i computes $G_i = \Phi_i^\top \Phi_i \in \mathbb{R}^{L\times L}$ and $H_i = \Phi_i^\top Y_i \in \mathbb{R}^{L\times 6}$. The server sums these statistics within each group and solves

$$\left(\sum_{i\in\mathcal{I}_g} G_i + \rho I_L \right) \widehat{\Theta}_g = \sum_{i\in\mathcal{I}_g} H_i, \tag{6}$$

where I_L is the $L \times L$ identity matrix. For a fixed partition, this gives the centralized solution in exact arithmetic without sharing Y_i. Positive regularization ensures a unique solution even with missing information. Identification from the data requires a separate rank condition.

Proposition 1: For a fixed nonempty group $\mathcal{I}_g$, let Φ_g be the vertical stack of its local designs Φ_i. The unregularized least-squares fit ($\rho = 0$) has a unique minimizer if and only if $\operatorname{rank}\Phi_g = L$, equivalently $\operatorname{rank}(\sum_{i\in\mathcal{I}_g} G_i) = L$. If, in addition, the targets satisfy $Y_i = \Phi_i\Theta_g$ for every group member, that minimizer equals Θ_g.

Proof: The aggregate Gram matrix is $\Phi_g^\top\Phi_g$, which is positive definite exactly when Φ_g has full column rank. Otherwise, a nonzero $U \in \mathbb{R}^{L\times 6}$ with $\Phi_g U = 0$ can be added to any minimizer without changing its predictions. For exact shared-map targets, Θ_g achieves zero residual and is therefore the unique minimizer in the full-rank case.

Pooling can therefore resolve the missing directions in Section II, but full rank does not remove the mismatch $d_i(v)$. Ridge supplies a solution, not missing data. The statement does not guarantee group discovery.

B. Assignment and model selection

Assignment balances the six matrix entries by scaling. For entry $j \in \{1, \dots, 6\}$, let σ_j be its standard deviation across all observed target rows, using the population divisor $\sum_i n_i$. The server computes these values from local row counts, entry sums, and squared sums. Define $s_j = \max\{\sigma_j, 0.05\,\mathrm{median}(\sigma_1, \dots, \sigma_6)\}$ and $D = \mathrm{diag}(s_1, \dots, s_6)$. These scales are positive in the evaluated fleets and fixed during each search. Given the broadcast maps, client i returns

$$q_{ig} = \|(Y_i - \Phi_i\widehat{\Theta}_g)D^{-1}\|_F^2, \quad g = 1, \dots, K, \tag{7}$$

and is assigned to the lowest-scoring group, with ties choosing the lowest index. Fitting still uses (5).

For initialization, fit all vehicles as one group to obtain the Global map $\widehat{\Theta}_0$. Client i sends its six-entry mean residual

$$a_i = \frac{1}{n_i} \sum_{\ell=1}^{n_i} \left[\widetilde{m}_i(v_{i,\ell}) - \phi(v_{i,\ell})^\top \widehat{\Theta}_0 \right]. \tag{8}$$

The server standardizes each entry across clients to zero mean and unit variance, uses unit scale for constant entries, and applies K-means. This scaling is separate from D. No complete local LPV map is needed.

Algorithm 1 summarizes the search. Define the residual sum $\mathrm{SSE} = \sum_{i=1}^{M} q_{i,g_i}$ and scalar target count $n = 6\sum_i n_i$. The Bayesian information criterion (BIC) score is

$$\mathrm{BIC} = n\log\left(\max\left\{\mathrm{SSE}/n, 10^{-12}\right\}\right) + p_K \log n, \tag{9}$$

Algorithm 1: Federated LPV Identification with Learned Groups

Input: Local designs Φ_i and targets Y_i, $i = 1, \dots, M$; common basis and regularization ρ.
Output: Selected K, assignments g_i, and maps $\widehat{\Theta}_g$, $g = 1, \dots, K$.
Prepare statistics and initialization summaries
Clients: send $G_i = \Phi_i^\top \Phi_i$, $H_i = \Phi_i^\top Y_i$, row counts, and target entry sums and squared sums
Server: fit Global $\widehat{\Theta}_0$ by (6) with all vehicles; compute D (Section III-B); broadcast both
Clients: send mean residuals a_i from (8)
Server: standardize the entries of a_i across clients
Search group counts and initial assignments
for $K = 1, \dots, 5$ **do**
  Use one start for $K = 1$ and 12 seeded starts otherwise
  for *each start* **do**
    Server: set $g_i = 1$ if $K = 1$; otherwise initialize by K-means on the standardized summaries
    for $t = 1, \dots, 30$ **do**
      Discard this start if any group has fewer than five vehicles
      Clients: send G_i, H_i and current group tags
      Server: fit each group by (6); broadcast all maps
      Clients: return scores q_{ig} from (7) for every group
      Server: assign each i to the lowest-scoring group, choosing the lowest index in a tie
      Break if no assignment changed
    Discard this start if any final group has fewer than five vehicles
    If the last iteration changed assignments, refit final groups and recompute scores without reassignment
    Server: compute BIC by (9); save this candidate
  Keep the lowest-BIC candidate for this K, if any
Select and distribute the result
Server: select the lowest-BIC candidate across K; return each vehicle's assignment and its group map

where $p_K = 6LK + K - 1$ counts map coefficients plus a nominal mixture-weight term. The procedure does not estimate mixture weights; BIC is used as a heuristic penalty for these noisy, heterogeneous targets. The lowest score selects the restart and K, including global pooling at $K = 1$. Neither global optimality nor physical-family recovery is guaranteed. Minimum group size does not ensure full scheduling rank, which is checked separately, not enforced during the search.

C. Exchanged information

The messages G_i and H_i contain $L(L+1)/2$ distinct Gram entries and $6L$ right-hand-side entries: 28 and 42 for $L = 7$. Communication accounting also includes row counts, target moments, initialization summaries, group tags, scores, and model broadcasts. Including all candidate group counts and restarts, mean label-free traffic is 2.243 MB per fleet at eight bytes per scalar. This exhaustive search is not shown to be communication-efficient. Records and matrix targets stay local in the distributed emulator, but the shared statistics can reveal information about them. No secure aggregation or differential privacy is implemented. The benchmark next tests identification beyond locally observed speeds.

TABLE I
VEHICLE PARAMETER CENTERS USED TO GENERATE THE FLEET. ALL CLIENTS HAVE $l_f = 1.2$ M AND $l_r = 1.6$ M.

Center	m [kg]	I_z [kg m^2]	C_f	C_r
Nominal	1500	2500	80000	80000
Heavy	1725	3000	84000	80000
Handling	1425	2250	72000	88000

Stiffnesses in N/rad.

IV. Benchmark and Evaluation

A. Nonlinear vehicles and calibration records

The plant is a two-state single-track model with nonlinear tanh tires:

$$\dot{\beta} = (F_f + F_r)/(mv) - r, \tag{10}$$
$$\dot{r} = (l_f F_f - l_r F_r)/I_z, \tag{11}$$
$$F_a = \mu F_{za} \tanh(C_a \alpha_a/(\mu F_{za})), \quad a \in \{f, r\}. \tag{12}$$

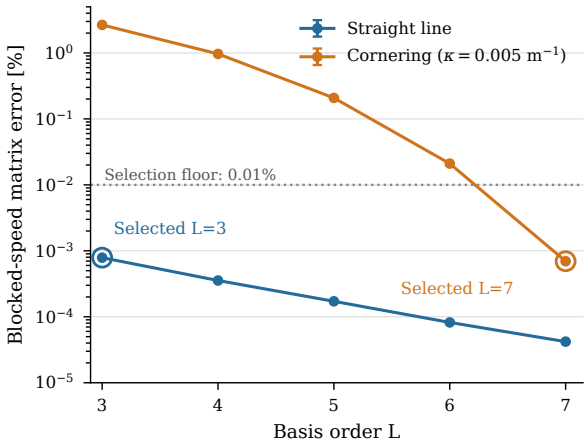
Here $\alpha_f = u - \beta - l_f r/v$, $\alpha_r = -\beta + l_r r/v$, $F_{zf} = mgl_r/(l_f + l_r)$, $F_{zr} = mgl_f/(l_f + l_r)$, $g = 9.81$ m/s^2, and $\mu = 0.9$. Each fleet has ten clients from each parameter center in Table I. Mass and inertia receive independent uniform multiplicative scatter of $\pm 2.5\%$ and axle stiffnesses $\pm 5\%$. Geometry remains fixed. Family labels are withheld from primary block allocation and fitting.

Calibration follows curvature $\kappa_0 = 0.005$ m^{-1}. At each observed speed, the simulator sets $r_i^\star = v\kappa_0$ and solves for $\beta_i^\star, u_i^\star$ with $\dot{\beta} = \dot{r} = 0$, using that client's plant parameters. Each of 36 records starts independently from $x_i^\star + \Delta x_j$, holds steering $u_i^\star + \Delta u_j$ and speed fixed for one fourth-order Runge–Kutta (RK4) step of $T_s = 0.01$ s, and supplies

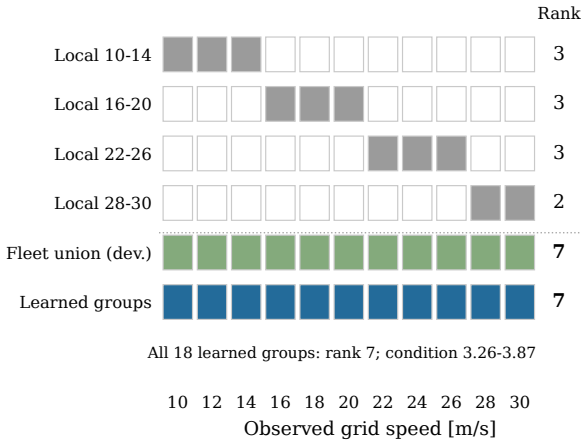
$$\begin{aligned} x_{j,+} &= \mathcal{F}_i^{\mathrm{RK4}}(x_i^\star + \Delta x_j, u_i^\star + \Delta u_j; v), \\ \widetilde{\Delta x}_{j,+} &= x_{j,+} - x_i^\star + \nu_j. \end{aligned} \tag{13}$$

The rows $[\Delta x_j^\top, \Delta u_j]$ and $\widetilde{\Delta x}_{j,+}^\top$ form $Z_i(v)$ and $X_{i,+}(v)$ in (2); the three-column regressor is checked for full rank. These records are independently reset transitions, not consecutive driving samples. In the primary study, only next-state targets receive noise ν_j. Zero-mean independent Gaussian perturbation standard deviations are approximately 0.25° for β, 2°/s for r, and 0.30° for steering. Added next-state noise has standard deviations 0.01° and 0.03°/s.

A prospective workflow is instrumented commissioning: hold selected speeds near steady cornering, estimate state/steering centers from settled data, and apply small steering perturbations with synchronized speed, steering, yaw-rate, and sideslip measurements. Sideslip requires a reference sensor or validated estimator. Local regression pairs supply the shared statistics. This acquisition procedure remains untested: real maneuvers cannot reset states arbitrarily, and correlated regressors, achievable excitation, and center estimation require validation. The sensitivity tests perturb the reset-record benchmark, not sequential driving or observer errors.



(a) Development-only basis validation



(b) Scheduling information ($L = 7$)

Fig. 2. Scheduling representation and information coverage. (a) Mean blocked-speed matrix error across ten development fleets; error bars denote one standard error. Rings mark the selected orders, and the dotted line is the 0.01% practical selection floor. (b) Filled cells mark observed grid speeds for the seven-function basis. The development union and all 18 audited primary groups have rank seven, unlike individual clients. Development union and learned-group coverage are distinct checks; full rank does not remove within-group bias.

The four local speed blocks are $\{10, 12, 14\}$, $\{16, 18, 20\}$, $\{22, 24, 26\}$, and $\{28, 30\}$ m/s. Globally balanced random allocation assigns them to eight, eight, seven, and seven clients, respectively, without consulting family labels. Primary fleet seeds are 371–380. Development representation selection uses separate seeds 301–310, contiguous blocked-speed validation, a one-standard-error rule, and a practical relative-error floor of 10^{-4}.

The target interval is $\mathcal{V} = [10, 30]$ m/s. Angles β, u are in radians and yaw rate r is in rad/s. The common scheduling basis uses $L = 7$ functions:

$$b(v) = [1, v^{-1}, v^{-2}, z, z^2, z^3, z^4]^\top, \quad z = (v - 20)/10. \tag{14}$$

The matrix $B_{\text{ref}} \in \mathbb{R}^{401\times 7}$ has rows $b(v)^\top$ on a 401-point uniform grid over $\mathcal{V}$. Its thin QR factorization is $B_{\text{ref}} = QR$, with $Q \in \mathbb{R}^{401\times 7}$ having orthonormal columns and invertible $R \in \mathbb{R}^{7\times 7}$. We use $\phi(v)^\top = b(v)^\top R^{-1}$, which changes conditioning without changing the represented span. Development-only blocked-speed validation selects $L = 7$ for cornering; the straight-line negative control retains $[1, v^{-1}, v^{-2}]$. Control comparisons do not select the representation. The primary study uses exact current state/input regressors and noisy next-state targets, with $N_i(v) = 36$ independently reset transitions per observed speed. Centering equilibria are supplied from simulator parameters. Section IV-C perturbs current measurements and assumed centers. Estimating centers from ordinary driving remains unvalidated.

B. Endpoints, averaging, and baselines

Reference matrices $m_i^\star(v)$ are numerical continuous-time Jacobians at the prescribed equilibrium, discretized by an augmented matrix exponential. They serve development-only representation and penalty selection and evaluation, but do not enter finite-data group fitting. Full-envelope matrix error is evaluated on the common speed grid:

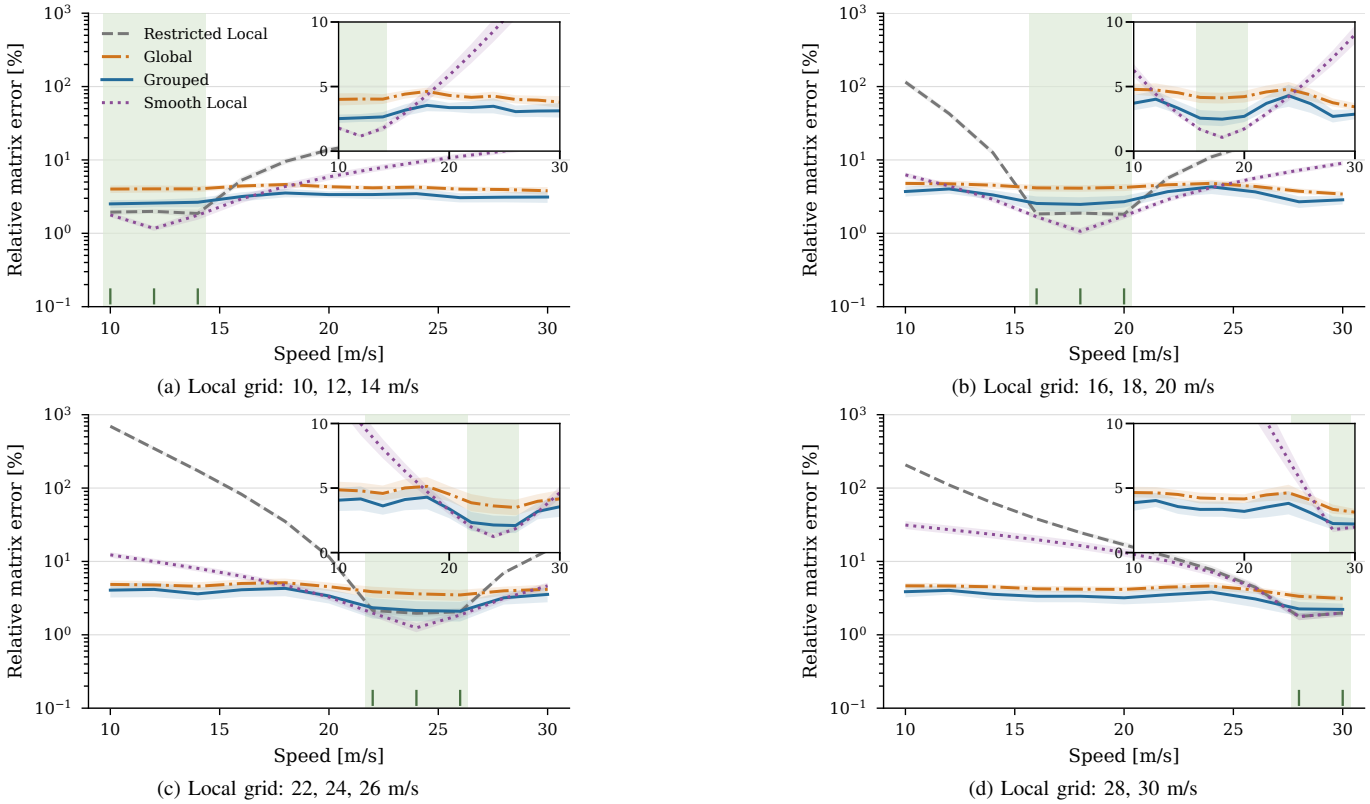
$$\begin{aligned} e_i(v) &= \frac{\|\widehat{m}_i(v) - m_i^\star(v)\|_2}{\|m_i^\star(v)\|_2}, \\ E_s &= \frac{1}{30 \cdot 11} \sum_i \sum_{v\in\mathcal{V}_{\text{grid}}} e_i(v). \end{aligned} \tag{15}$$

Here $\mathcal{V}_{\text{grid}} = \{10, 12, \ldots, 30\}$ m/s and s indexes fleets. The metric uses the stated physical coordinates without entry normalization. It is coordinate dependent and does not directly measure tracking performance. A post-hoc check averages

$$e_i^{\text{sc}}(v) = \sqrt{\frac{1}{6}\sum_{j=1}^{6}\left(\frac{\widehat{m}_{ij}(v) - m_{ij}^\star(v)}{d_j}\right)^2}$$

over the same pairs. Each d_j is the RMS reference entry across development fleets 321–330 and all eleven speeds. These fixed scales balance entries, without defining a control-weighted norm. Entry diagnostics use mean absolute error (MAE). Observed/unobserved-speed summaries average only their respective client-speed pairs within each fleet. Reported errors average E_s over ten independent fleets. For method M and baseline B, reported gains average $100(1 - E_s^M/E_s^B)$ over paired fleets, rather than taking the ratio of grand mean errors. Confidence intervals resample those ten fleet units 20,000 times.

Restricted Local uses $L = 3$, ridge 10^{-8}, and each client's records. Global and both grouped methods use $L = 7$ and the same fleet records. These baselines separate complementary coverage, compatibility selection, and distributed implementation. Prior-centered Local uses $L = 7$ with ridge toward its own $L = 3$ fit expressed in the richer basis, denoted $\Theta_{i,0}$. Leave-one-speed-out validation selects the penalty ρ_i from $\{10^{-6}, 10^{-4}, 10^{-2}, 1, 10^{2}\}$; clients with only two speeds use the fixed middle candidate 10^{-2}.



(a) Local grid: 10, 12, 14 m/s

(b) Local grid: 16, 18, 20 m/s

(c) Local grid: 22, 24, 26 m/s

(d) Local grid: 28, 30 m/s

Fig. 3. Primary matrix error by local speed block, with family labels withheld. Light shading marks each observed block; short green marks identify its observed grid speeds. Curves average clients within block and fleet, then across ten fleets; bands are pointwise 95% fleet-bootstrap intervals from 20,000 draws. The logarithmic axis retains Restricted Local extrapolation failure. Insets show Global, Grouped, and Smooth Local on a linear 0–10% scale with identical averaging and bands. Smooth Local uses the frozen curvature prior.

To explain its near-equivalence to Restricted Local, define $R_i = Y_i - \Phi_i\Theta_{i,0}$, $\Delta_i = \widehat{\Theta}_i^{\mathrm{P}} - \Theta_{i,0}$, and $P_i = \Phi_i^\top\Phi_i + (\rho_i+\epsilon)I$. The centered-ridge normal equations imply

$$\begin{aligned} P_i\Delta_i &= \Phi_i^\top R_i - \epsilon\Theta_{i,0}, \\ N_i\Delta_i &= -\frac{\epsilon}{\rho_i+\epsilon} N_i\Theta_{i,0}, \end{aligned} \tag{16}$$

where N_i has orthonormal rows spanning $\ker\Phi_i$ and $\epsilon = 10^{-12}$. If $R_i = 0$ and $\epsilon = 0$, then $\Delta_i = 0$ for every $\rho_i > 0$. The three-function design interpolates two or three distinct-speed targets; ridge gives near-interpolation. The richer fit therefore retains the prior's unsupported directions apart from numerical shrinkage.

To test sharing against a distinct local prior, Smooth Local penalizes function curvature with $L = 7$:

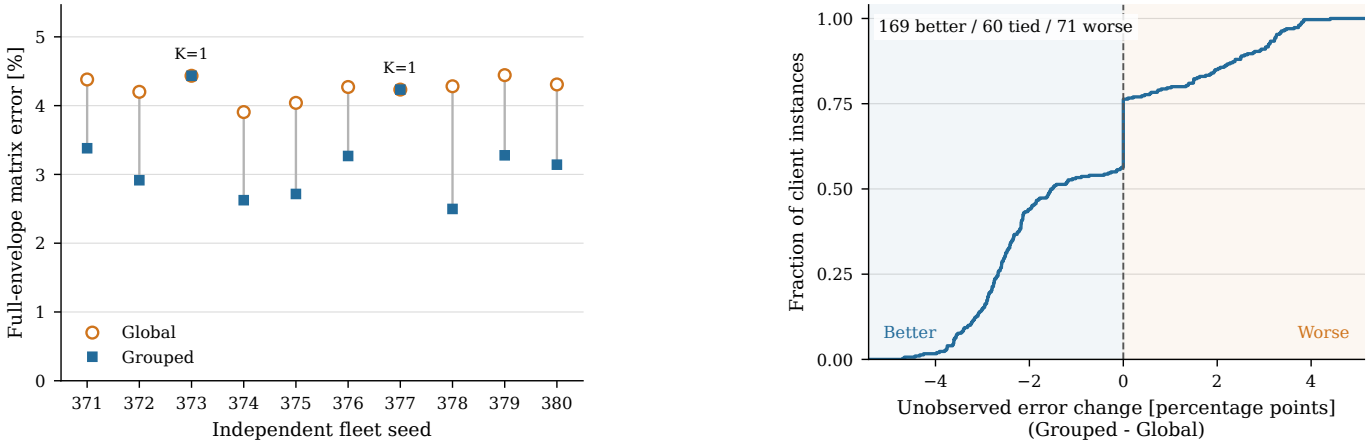
$$\begin{aligned} \widehat{\Theta}_i^{\mathrm{S}} = \arg\min_{\Theta}\Big\{ &\frac{\|Y_i - \Phi_i\Theta\|_F^2}{n_i} \\ &+ \alpha\|Q_2\Theta\|_F^2 + \eta\|\Theta\|_F^2\Big\}. \end{aligned} \tag{17}$$

Here $\|Q_2\Theta\|_F^2$ uses 401-point trapezoidal quadrature for $\frac{1}{2}\int_{-1}^{1}\|d^2[\phi(v(z))^\top\Theta]/dz^2\|_2^2\,dz$; $\eta = 10^{-16}$ is numerical ridge. The curvature penalty leaves $\mathrm{span}\{1, z\}$ unpenalized, distinct from Restricted Local's $\mathrm{span}\{1, v^{-1}, v^{-2}\}$. Smoothness supplies a prior, without restoring local scheduling rank. A common α is selected from $\{10^{-6}, 10^{-4}, 10^{-2}, 1, 10^2, 10^4, 10^6\}$ on development fleets 321–330 using unobserved-speed matrix error. The largest penalty within one fleet standard error of the best mean is chosen: 10^6, approaching an affine speed map. Reference Jacobians serve this offline tuning; recipient fitting uses only its own noisy targets at the original two or three speeds. This matched check and the group-information audit reuse the ten primary fleets and their saved calibration targets. They provide no independent confirmation. A full-coverage recipient reference outperforms family pooling at approximately equal transition budget, bounding the benefit to missing local access rather than superiority at equal recipient information.

C. Focused robustness protocol

To test the dependence on pooled coverage and calibration quality, ten fresh fleets (seeds 1201–1210) are evaluated under seven predeclared paired conditions. The reference repeats the primary procedure. All conditions retain each client's two- or three-speed count, 36 transitions per speed, physical parameters, basis, grouping settings, and the frozen Smooth Local penalty. Perturbation and noise draws are paired by client and speed-slot index; coverage changes alter speed locations while retaining record budgets.

For family-coverage restriction, all ten heavy clients observe the first two or three speeds in $\{10, 12, 14\}$ m/s; other clients retain their blocks. Labels enter only this stress generator



(a) Paired primary results

(b) Client tradeoff (descriptive)

Fig. 4. Primary fleet gains and client tradeoffs. (a) Paired full-envelope errors for all ten independent fleets; connectors join methods within each fleet, and $K = 1$ marks the two ties. (b) Descriptive empirical cumulative distribution of client mean unobserved-speed error changes (Grouped minus Global) in percentage points; negative values indicate improvement. Ties use a 10^{-12} dimensionless-error tolerance. The 300 simulated clients are nested within ten fleets.

and a separate OracleFamily diagnostic, never learned fitting or selection. Fleet-coverage restriction maps the four blocks to $\{10, 12, 14\}$, $\{16, 18, 20\}$, $\{10, 12, 14\}$, and $\{16, 18\}$ m/s, respectively. Four calibration conditions separately halve state/input excitation standard deviations, double next-state noise standard deviations, add current measurement noise with standard deviations $(0.01°, 0.03°/\mathrm{s}, 0.01°)$ for (β, r, u), or bias their assumed centers by magnitudes $(0.05°, 0.2°/\mathrm{s}, 0.1°)$. Bias signs are drawn independently per component/client and held fixed across its speeds. Center bias changes only the subtraction in (2), retaining identical measured records and zero-intercept OLS.

A predeclared rank-only fallback replaces deficient selected group maps with Global after selection, without changing grouping or fitting. Global deficiency is flagged separately. The fixed full-grid error (15) is the main endpoint; recipient-unobserved sets change with coverage. Intervals are pointwise 95% paired-fleet bootstrap intervals. No representation, prior, or controller is retuned, and no tracking rollout is added.

V. Identification Results

A. Why complementary coverage is needed

At curvature 0.005 m^{-1}, development blocked-speed matrix error falls from 2.6607% at $L = 3$ to 0.0007% at $L = 7$. Figure 2(a) contrasts this basis requirement with the straight-line case, which retains order three. For seven coefficients, the local blocks have ranks three, three, three, and two; the development union has rank seven and condition 3.406 (Fig. 2(b)). Complementary coverage can therefore supply missing directions, conditional on the learned partition.

With finite transitions and known compatibility, oracle-family pooling achieves 1.816% unobserved-speed error, versus 66.121% for Restricted Local and 4.296% for Global on separate development fleets. The prior-centered diagnostic changes Restricted Local negligibly there. These oracle results motivate the mechanism; the primary evaluation below withholds labels.

B. Primary label-free results

Table II separates the full-envelope endpoint from its observed and unobserved regions. Mean paired gains over Restricted Local and Global are 93.41% and 23.74%, respectively. The Global-relative 95% interval is $[15.39, 31.04]\%$, with improvement in eight of ten fleets. Two groups are selected in eight fleets and one in two. Local methods have lower mean observed-speed error than grouped fitting under the primary metric (Table II). Sharing substantially reduces Restricted Local's unsupported extrapolation error. Grouped sharing additionally reduces Global's pooling error. Figure 3 locates these changes relative to each client block. Figure 4(a) shows the paired fleet errors, including the two $K = 1$ ties. Aggregate improvement does not extend to every client: 169 clients improve, 60 are numerical ties, and 71 have larger unobserved-speed error than Global. Figure 4(b) shows this tradeoff. The 300 simulated clients are nested within ten fleets; confidence intervals resample fleets, not clients.

Prior-centered Local changes unobserved-speed error by only 0.00016 percentage points. Its prior's observed-target relative residual has median 0.0020% and maximum 0.0708% across 300 clients. Equation 16 explains the prior retention; its correction and nullspace identities are verified for all 300 saved fits.

Smooth Local reduces Restricted Local's extrapolation error and has the lowest observed-speed error (Table II). Against Smooth Local, grouped fitting gives mean paired gains of 56.20% in full-envelope error and 62.33% in unobserved-speed error. The 95% fleet-bootstrap intervals are $[49.69, 61.81]\%$ and $[57.03, 67.08]\%$, respectively. Both endpoints improve in all ten fleets. Figure 3 shows the remaining unobserved-speed gap. Other local priors remain untested.

The fixed-scale diagnostic retains mean paired gains over Global of 21.28% full-envelope and 17.93% unobserved, with pointwise fleet-bootstrap intervals $[13.65, 28.11]\%$ and $[10.94, 24.68]\%$. Entry-wise improvement is not uniform:

TABLE II

MEAN MATRIX ERROR [%] ACROSS TEN PRIMARY FLEETS WITH LABEL-FREE GROUPING. PRIOR-CENTERED LOCAL RETAINS THE RESTRICTED PRIOR; SMOOTH LOCAL TESTS FUNCTION CURVATURE.

Method	Full	Observed	Unobserved
Restricted Local	49.562	1.930	65.568
Prior-centered Local	49.562	1.930	65.568
Smooth Local	7.489	1.630	9.458
Global	4.250	3.840	4.388
Centralized groups	3.249	2.423	3.527
Federated groups	3.249	2.423	3.527

unobserved B_1 MAE increases by 30.90% on average and deteriorates in every fleet; the A_{12} gain interval includes zero. This reuses primary fits and does not independently confirm them.

Federated and centralized protocols select identical partitions and group counts in all 20 fleet/protocol jobs, including the separate family-stratified coverage reference. Maximum coefficient discrepancy is 3.20×10^{-14} and maximum BIC discrepancy is 1.87×10^{-9}. This verifies the implementation under matched inputs.

C. Scheduling information in selected groups

All 18 primary selected groups contain observations at all eleven grid speeds and have scheduling-design rank seven. Their stacked design condition numbers range from 3.26 to 3.87 in the fixed QR basis, as summarized in Fig. 2(b). The learned partitions satisfy Proposition 1's scheduling condition for this benchmark. The reported conditioning concerns the scheduling design, rather than its Gram or the full dynamical-identification information matrix. Full rank neither removes within-group bias nor guarantees physical-family recovery.

Retrospective family diagnostics clarify this distinction. In all eight $K = 2$ fleets, heavy and handling clients occupy separate groups, while nominal clients split between them. Labels remain withheld from learning. All 71 clients with larger unobserved error than Global are nominal; their family mean worsens in each $K = 2$ fleet. Thus compatible sharing is an aggregate regression compromise with sufficient coverage, rather than three-family recovery or uniform recipient benefit. These patterns do not establish why BIC selects two groups.

D. Coverage and calibration robustness

Table III reports the robustness cohort separately from the primary fleets. Restricting heavy-family coverage leaves its ten-client OracleFamily design at rank three, despite exceeding the minimum size of five. Its family-specific full-grid error rises from 1.860% to 69.225%. Thus client count and known compatibility do not ensure scheduling information. The fleet and all selected learned groups remain rank seven. Each heavy-client group includes clients from other families. Federated fitting retains a mean paired 15.49% gain over Global, interval $[5.03, 26.76]\%$. This observed pooling response is not a rank guarantee from BIC.

TABLE III

ROBUSTNESS STUDY: MEAN FULL-GRID MATRIX ERROR [%] ACROSS TEN PAIRED FLEETS. n_1 IS THE NUMBER OF FLEETS SELECTING ONE GROUP ($K = 1$); THE OTHERS SELECT $K = 2$. SMOOTH DENOTES THE FROZEN SMOOTH LOCAL PRIOR.

Condition	Federated	Global	Smooth	n_1
Reference	3.058	4.214	7.221	3
Family restriction	3.879	4.584	7.779	5
Fleet restriction	33.423	33.255	6.645	5
Half excitation	4.324	4.324	13.179	10
Double target noise	4.326	4.326	13.286	10
Noisy regressors	4.039	4.258	9.707	7
Biased centers	4.810	4.906	9.734	9

Fleet-wide restriction instead gives rank six for Global and every selected or oracle-family group. Federated error reaches 33.423%, versus 6.645% for Smooth Local. On the five speeds absent from the entire fleet (22–30 m/s), their errors are 69.302% and 10.437%. The Global fallback is itself deficient and cannot supply the missing direction. Smoothness provides useful extrapolation in this condition, without restoring identifiability. An isolated deficient selected group with full-rank Global does not occur in this study; rescue of that case by the fallback is therefore not demonstrated.

Weaker excitation and doubled target noise select $K = 1$ in all ten fleets, making Federated identical to Global. Both reduce calibration signal-to-noise and give similar errors under the paired perturbations. Noisy regressors and biased centers also reduce the selected group count; their paired gain intervals over Global include zero. Sharing remains more accurate than Smooth Local in all four calibration conditions, but a grouping advantage over Global is not established there. All local transition designs retain rank three, and selected scheduling designs remain rank seven in these conditions: calibration sensitivity and scheduling deficiency are distinct limits. These are controlled identification tests, not online sensor-noise or tracking-robustness validation.

VI. SUPPORTING LQI VALIDATION

Ten separate confirmation fleets (seeds 411–420) refit the maps and apply speed-scheduled LQI to nonlinear 12 s maneuvers with available noise-free physical-state feedback. Each fleet repeats the partial-speed calibration above; maps remain fixed during tracking. ExactLPV is a recipient-specific $L = 7$ fit to exact Jacobians on the eleven-speed grid. At each grid node, the method's fitted $\widehat{A}, \widehat{B}$ define

$$A_{\mathrm{a}} = \begin{bmatrix} \widehat{A} & 0 \\ T_s c_r & 1 \end{bmatrix}, \quad B_{\mathrm{a}} = \begin{bmatrix} \widehat{B} \\ 0 \end{bmatrix}, \quad c_r = [0, 1]. \tag{18}$$

Discrete Riccati synthesis uses $Q = \mathrm{diag}(25, 50, 1000)$ for $[\beta, r, \xi]^\top$, with yaw-error integral ξ, and steering weight $R = 0.15$. The yaw-reference prefilter is $F = w + K_x h$, where $(I - \widehat{A})h - \widehat{B}w = 0$ and $c_r h = 1$; K_x denotes the first two entries of the LQI gain. For Restricted Local at unobserved grid nodes, synthesis evaluates its matrix map at the nearest observed speed. All methods then linearly interpolate gains

TABLE IV

SUPPORTING YAW TRACKING RMSE [DEGREES/S], AVERAGED ACROSS TEN CONFIRMATION FLEETS. VIOLATIONS COUNT CLIENTS EXCEEDING THE 6° SIDESLIP BOUND (MODERATE/HARD), OUT OF 300 EACH. SMOOTH LOCAL MEANS INCLUDE FAILED RESPONSES. RESTRICTED LOCAL USES THE NEAREST OBSERVED SPEED OUTSIDE ITS BLOCK.

Method	Moderate	Hard	Violations
Restricted Local	0.0540	0.1698	0/0
Smooth Local	8.3565	5.7161	88/65
Global	0.0429	0.1564	0/0
Federated groups	0.0411	0.1546	0/0
ExactLPV reference	0.0400	0.1507	0/0

and prefilters on the common grid. Weights were selected using only ExactLPV performance on development seeds 381–385, subject to feasibility and saturation checks. A weaker maneuver pilot on seeds 391–400 was discarded before the final confirmation. Neither grouped-versus-Global results nor confirmation fleets select weights. Smooth Local uses the same fleets and targets, frozen $\alpha = 10^6$, and all grid nodes. The common controller protocol remains frozen; this is neither a separately tuned comparison nor an independent cohort.

With $r_k^{\mathrm{ref}} = a_{y,k}^{\mathrm{req}}/v_k$, the runtime law is

$$\begin{aligned} u_k^{\circ} &= -K(v_k)[\beta_k, r_k, \xi_k]^{\top} + F(v_k) r_k^{\mathrm{ref}}, \\ \xi_{k+1}^{\mathrm{cand}} &= \mathrm{clip}_{[-0.6,0.6]}\{\xi_k + T_s(r_k - r_k^{\mathrm{ref}})\}. \end{aligned} \tag{19}$$

Steering is rate-limited relative to u_{k-1}, then amplitude-limited. The integral candidate is accepted when the command is effectively unsaturated or the integral increment counteracts limiting; otherwise $\xi_{k+1} = \xi_k$. The integral bound is in radians. Each maneuver starts with $x_0 = 0$, $\xi_0 = 0$, and $u_{-1} = 0$; RK4 holds the applied steering and current speed fixed within each sample.

Feedback acts on absolute states, with reference feedforward from the homogeneous fitted matrices. Calibration equilibrium offsets are not used online. The maps describe one cornering curve, whereas the maneuvers vary the requested turn; tracking therefore tests empirical usability beyond that curve, rather than exact equilibrium-offset compensation.

The moderate maneuver smoothly varies speed $11 \to 29 \to 15$ m/s and uses an enveloped two-frequency lateral-acceleration demand with scale 4 m/s^2 and frequencies 0.35 and 0.72 Hz. The hard maneuver varies speed $29 \to 12 \to 28$ m/s and uses alternating Gaussian demands centered at $3.0, 6.0, 9.2$ s, with respective widths $0.40, 0.48, 0.55$ s and amplitudes $7, -7.7, 5.6$ m/s^2. Yaw reference is requested acceleration divided by speed. Steering limits are 12° and 120°/s, with a 6° sideslip feasibility bound. These remain fixed across methods. Yaw RMSE uses the complete maneuver, then averages clients within fleet and the ten fleet means.

For moderate and hard demands, mean paired tracking gains are 23.86% and 8.96% against Restricted Local, and 4.26% and 1.21% against Global, respectively. Global-relative intervals are $[1.95, 6.55]\%$ and $[0.47, 2.01]\%$. Absolute mean improvements are 0.001824 and 0.001874 degrees/s, so the control evidence supports usability with modest practical headroom. All federated trajectories satisfy the tested constraints. The maximum frozen-point spectral radius is 0.9759, evaluated on 81 speeds using the interpolated gain and ExactLPV reference augmentation (18). This unsaturated diagnostic on the calibration curve is not a scheduled or nonlinear stability certificate. Four fleets select one group, making federated and Global controllers identical there.

Smooth Local separates matrix accuracy from controller suitability. All 300 schedules synthesize and 600 rollouts finish numerically, but $114/300$ clients have frozen-point radius at least one (maximum 50.81). Sideslip violations affect $88/300$ moderate and $65/300$ hard runs. Severe tails dominate Table IV; client median RMSEs are 0.0589 and 0.2047 degrees/s. Bound violations mark failed responses, not credible vehicle behavior. This prior is unreliable under the frozen protocol; other priors and separately tuned controllers remain untested.

Using archived original profiles, vehicle parameters, and saved schedules, all 600 Smooth Local trajectories and tracking metrics reproduce bitwise across the tested platforms. Regenerated profiles differed in speed by 3.55×10^{-15} m/s, amplifying numerical differences. Table IV retains the original results.

VII. DISCUSSION AND CONCLUSION

Complementary speed blocks supply missing local directions, while grouping reduces aggregate pooling error relative to Global. The primary advantage persists under development-scaled entry weighting, but does not extend to every entry or recipient. Heavy and handling clients benefit while many nominal clients deteriorate; learned groups need not recover physical families. Centralized agreement verifies the distributed implementation.

The robustness study bounds the benefit: minimum group size does not ensure rank, and Global fallback cannot restore fleet-wide missing information. Smooth Local wins in that deficient condition; weaker calibration signal-to-noise can collapse grouping to Global. Full selected-group rank is empirical, selection is heuristic, and exhaustive search incurs measurable communication. Ordinary driving, observer errors, real fleets, and frozen-library transfer remain unvalidated.

The scheduled-LQI tests support usability with modest gains over Global. The tested Smooth Local prior yields infeasible responses for some clients under the frozen protocol, illustrating that matrix accuracy alone does not establish controller suitability. Compatible sharing is useful when pooled coverage and calibration quality support it, with explicit recipient tradeoffs.

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